Problem Set 11

Submission:

Friday, 15/05/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage.

Note: This problem set is shorter, so that the mandatory problems can in principle be solved until the end of the semester. Nevertheless, the deadline is 15/05/2026.

Only problems marked with $(*)$ will be graded.

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1. $(*)$ **Convergence in probability** [4 Points]

   Let $(\Omega, \mathscr{F}, \mathbf{P})$ a probability space. Suppose that for every $n \in \mathbb{N}$, we have random variables $X_n, Y_n : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$, and let $X, Y : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$ be random variables.

   (a) Suppose that $X_n \xrightarrow[n\to\infty]{\mathbf{P}} X$, $Y_n \xrightarrow[n\to\infty]{\mathbf{P}} Y$. Show that
   $$X_n + Y_n \xrightarrow[n\to\infty]{\mathbf{P}} X + Y.$$

   (b) Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of real numbers with $\lim_{n\to\infty} a_n = a$. Suppose that $X_n \xrightarrow[n\to\infty]{\mathbf{P}} x$, for some $x \in \mathbb{R}$. Show that
   $$a_n \cdot X_n \xrightarrow[n\to\infty]{\mathbf{P}} a \cdot x.$$

2. **Weak law of large numbers**

   (a) Let $(X_n)_{n\in\mathbb{N}}$ be i.i.d. real random variables with $X_1 \sim \mathscr{E}(\frac{1}{2})$. Use the weak law of large numbers to show that
   $$\frac{1}{n}\sum_{i=1}^{n} \exp\left(-\frac{X_i}{2}\right) \xrightarrow[n\to\infty]{\mathbf{P}} \frac{1}{2}.$$

   (b) Let $(X_n)_{n\in\mathbb{N}}$ be i.i.d. real random variables such that the law of X_1 has the probability density function
   $$f_{X_1}(x) = \begin{cases} \frac{1}{6}, & 0 \leq x < 2, \\ \frac{1}{3}, & 2 \leq x \leq 4, \\ 0, & \text{else} \end{cases}$$
   Calculate the value of $a \in \mathbb{R}$ such that
   $$\frac{1}{n}\sum_{k=1}^{n} \sqrt{X_k} \xrightarrow[n\to\infty]{\mathbf{P}} a.$$

3. **Applications of the Borel-Cantelli lemmas**

   (a) Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of i.i.d. $\mathscr{E}(\lambda)$-distributed random variables, with $\lambda > 0$.

      i. Show that
      $$\limsup_{n\to\infty} \frac{X_n}{\log(n)} = \frac{1}{\lambda}, \qquad \mathbf{P}\text{-a.s.}$$

      ii. Show that
      $$\liminf_{n\to\infty} \frac{X_n}{\log(n)} = 0, \qquad \mathbf{P}\text{-a.s.}$$

(b) Consider $X_1, X_2, X_3, \dots$ independent random variables such that $\mathbf{P}[X_1 = 0] = 1$, and

$$\mathbf{P}[X_n = n] = \mathbf{P}[X_n = -n] = \frac{1}{2n\log(n)}, \qquad \mathbf{P}[X_n = 0] = 1 - \frac{1}{n\log(n)},\ n \geq 2.$$

Show that $\frac{1}{n}\sum_{i=1}^{n} X_i$ converges to 0 in $\mathbf{P}$-probability, but not $\mathbf{P}$-a.s..

4. $(*)$ **Convergence in distribution** [4 Points]

Consider sequences of real random variables $(X_n)_{n\in\mathbb{N}}$ and $(Y_n)_{n\in\mathbb{N}}$ with probability density functions

(i) $f_{X_n}(x) = \lambda\left(1 - \frac{\lambda x}{n}\right)^{n-1} \mathbb{1}_{[0,\frac{n}{\lambda})}(x)$, where $\lambda > 0$,

(ii) $f_{Y_n}(x) = \frac{n+1}{n} x^{\frac{1}{n}} \mathbb{1}_{[0,1]}(x)$,

Show that X_n and Y_n converge in distribution as $n \to \infty$, and determine the respective limit distribution.